

Narrative Package V2 - Colony Horizon

Project: Dark Matter: Genesis

Package identity: V2 Colony Horizon

Player: Kade

Tone: Hope / colony-builder / competent survival / earned legacy

Designed hour target: 170 hours excluding exploration, free roam, harvesting drift, build decoration, and post-ending sandbox

Canon locks: Io 2160, Aether Credits only, starter **5000 AC**, 22 base companions + expedition trio, Kairos repair -> 6 Memory Cores -> trust, biome order **B6 -> B1 -> B2 -> B3 -> B5 -> B4 -> B7**, Echo fail permanent, base injury-not-death.

1. Premise, Tone, And Kairos Truth

Kade does not come to Io to win a war against the moon. Kade comes to prove that people can build a future where every previous expedition failed.

The first colony horizon is a fragile line: a pressure shell, a storm shelter, a Building Control Panel that barely boots, and a roster of survivors who need reasons to keep working. The fantasy is not domination. It is **construction under schedule pressure**: raising habs before sulfur storms, routing power through colony Barrier Fields, restoring production queues, rescuing Echoes before sync collapse, and watching a hostile world become legible one building at a time.

Kairos is not an enemy AI. It is a lost expedition archive. Its Memory Cores are not loot containers or superweapons; they are people, crew imprints, field journals, unfinished promises, and names preserved badly by a machine that has been alone too long. Repairing Kairos wakes a grieving witness. Restoring all six cores gives the colony enough truth to choose what kind of future it deserves.

Endings:

- **Build:** Ratify a colony charter and remain on Io. Kairos becomes the Colony Voice, an advisor bound by public trust.
- **Evacuate:** Leave with the restored cores and names. Io stays dangerous, but the expedition stops being erased.
- **Seed:** Plant a hybrid Echo-colony legacy. The colony accepts ongoing Resonance/Saturation costs to create something neither Earth nor old Io intended.

Hopeful dialogue rule: Even fear lines should point toward agency: shelter, repair, witness, vote, seed, carry names forward.

2. Biome And Ecology Integration Map

2.1 Biome Progression

Code	Biome	Role In V2 Colony Horizon	Ecology And Encounter Use	Content Status
B6	Basalt Highlands	Hub, landing zone, first BCP shell, first colony ring, companion base.	Cliff Tube Lace, Tube Jackal, Brood Tunnel Mouth, Glass Hive Swarmer, Cave Scout Moth, Beacon Hopper; Brimstone Puff pet migration hook.	[PLANNED] biome, [SHIPPED] BCP shell hooks
B1	Sulfur Plains	First storm shelter pressure, Core 1, proof that base injury is not death.	Sulfur Hound [SHIPPED], Brimstone Fan, Cinder Skitter, Brimstone Leech, Graveyard Scrapper Drone, sulfur storm behavior, B1 ecology table [SHIPPED].	[SHIPPED] Sulfur Hound + B1 table, [PARTIAL] weather stubs
B2	Geyser Fields	Greenhouse repairs, vent timing, C2 fertility systems.	Vent Crab Worker/Queen, Plume Moth, Geyser Pod, Geyser Strider, Vent Capper Bot, Rusted Survey Drone.	[PLANNED] biome, [PARTIAL] hovercraft traversal
B3	Ash Flats & Ridges	Waystations, beacons, Barrier Field routing, caravan weather schedules.	Basalt Jackal, Dust Spout Cluster, Ash Stalker, Ash Glass Wasp, Salvage Excavator Android, Claim Jumpers (H6).	[PLANNED] biome
B5	Polar Radiation Flats	Radiation/cold survival, Core 4, Void ecology, Freeboot Relief arrival.	Void Kelp, Magnet Wyrms, Rift Stalker, Cold Spire Hound, Smuggler Remnant Android, Mag-Clamp Drone.	[PLANNED] biome, [SHIPPED] exposure base
B4	Lava Calderas	Heat survival, memorial forge, Core 5, geothermal modules.	Caldera Mantis, Heat Eel, Magma Skitter, Caldera Heat Kite, Eruption Sentry Bot, Rim Glass Needle Mat.	[PLANNED] biome
B7	Precursor Ruin Belt	Final archive, C6 charter vault, ending triad.	Vault Stalker, Silence Moth, Still Hunter (myth), Resonance Echo Shelf, Rust Garden, Corrupted Patrol Android, Void Stitcher (global).	[PLANNED] biome

2.2 Underground Strata

Stratum	Name	Use	Ecology
UG Stratum 1	Upper Lava Tubes	First cellar routes and BCP utility runs.	Tube Lace, Tube Jackal, Cinder Tunnel Skitter, Rust Garden.
UG Stratum 2	Mid Galleries	Gas dome and greenhouse support puzzles.	Glass Kelp, Rift Skimmer, Glass Hive Swarmer, Brood Tunnel Warden.
UG Stratum 3	Deep Volatile Basins	Risk/reward farms and survival puzzles with no hard gate.	Brine Fan, Basin Mantis, Brine Hound, Lamprey Spire Colony.
UG Stratum 4	Geothermal Roots	Combat setpieces and heat loot.	Heat Eel, Magma Phase Crawler, Silicate Mirror Bloom.
UG Stratum 5	Resonance Vaults	B7 precursor support route and Seed ending prep.	Echo Lichen, Vault Stalker, Echo Symbiont Swarm, Corrupted Patrol Android.

2.3 Required Organism Roster

Use the real ecology names in quests, puzzles, combat tables, life sheets, and codex text:

- **Shipped ecology:** Sulfur Hound [SHIPPED].
- **Surface fauna:** Vent Crabs, Plume Moth, Basalt Jackal, Dust Spout, Caldera Mantis, Heat Eel, Void Kelp, Magnet Wyrms, Rift Stalker.
- **Underground fauna:** Tube Jackal, Brood Tunnel, Glass Hive, Vault Stalker, Silence Moth, Still Hunter, Void Stitcher, Rust Garden.
- **Synthetic families:** androids **A1-A10**, humanoids **H1-H7**, machines **M1-M7**, flyers **F1-F8**.
- **Pets:** Brimstone Puff + core12 pet roster [PLANNED], folded into the Echo/trio framework as migration work.

2.4 Life Sheet References

Durable art/design references (read-only — never delete/move):

Sheet	Path
Manifest	Assets/_Project/Documentation/Design/Io_Biome_Life_Sheet_Manifest.md
Plan	Assets/_Project/Documentation/Design/Io_Biome_Life_Image_Sheet_Plan.md
Set A PBR	Assets/_Project/Documentation/Design/ArtReference/LifeSheets/
Set B RT	Assets/_Project/Documentation/Design/ArtReference/LifeSheets_RayTraced/ (RT_LifeSheet_B1_B2_Sulfur_Geyser.png, ...B3_B4_Ash_Caldera.png, ...B5_B6_Polar_Highlands.png, ... B7_Ruins_Global.png, underground + threats + pets sheets)

V2 uses them as ecology vocabulary and encounter composition references.

3. Systems Inventory Hooked By Content

System	Status	V2 Content Hook
Aether Credits (AC)	[SHIPPED]	Starter 5000 AC ; all quest, puzzle, vendor, upgrade, and faction rewards use AC only.
Building Control Panel shell	[SHIPPED]	Every major colony structure has an in-world E terminal with Overview, Companions, Production, Craft, Changes tabs.
BCP Craft / Changes tabs	[PARTIAL]	V2 quests teach production queues, module changes, and repair schedules through BCP tasks.
EchoGenerator / chronicle	[SHIPPED]	Echo names, core memories, trust dialogue, and permanent fail stakes.
Exposure	[SHIPPED]	Cold/heat, radiation, sulfur storms, shelter timing, B5/B4 survival planning.
Sulfur Hound + B1 table	[SHIPPED]	First reliable predator table; used in B1 storm escort and shelter perimeter encounters.
World Engine spine	[SHIPPED]	World -> Simulation -> Intelligence -> Experience -> Presentation -> Player supports triggered mainline beats.
Combat / hotbar	[SHIPPED]	Named weapons map to current melee/ranged/hotbar behaviors where possible.
Weather stubs	[PARTIAL]	V2 treats weather as schedule: storm building, peak, dissipating, clear windows.
Hovercraft	[PARTIAL]	Used as later logistics/traversal fiction, not required for Act I gates.

System	Status	V2 Content Hook
Quest scaffold	[PARTIAL]	Mainline advances via progress triggers; sides/factions can use NPCs and boards.
Pets legacy	[PARTIAL]	Brimstone Puff and core12 migrate into Echo/trio companion design.
Kairos / cores / Resonance	[PLANNED]	Six Memory Cores restore people and trigger settlement-scale events.
Communications	[PLANNED]	Beacon Homestead, Scout Relay Mesh, and later colony scheduling.
Purification Hub live	[PLANNED]	Saturation cleansing, Hearth Clinic, Seed ending cost management.
Biomes	[PLANNED]	V2 content assumes biome implementation in locked order.
Underground	[PLANNED]	UG Strata 1-5 support farms, caches, combat hives, and B7 archive routes.
Modules	[PLANNED]	Generators, power grids, automated gather/logistics, communications, defense, mining.
Pet migration	[PLANNED]	Fold pets into Echo/trio roster instead of separate pet loop.
Weather Director	[PLANNED]	Required to make weather windows, storm shelter mandates, and scheduled colony fantasy live.

4. AC, Weapons, And Upgrade Curve

4.1 AC Scale Curve

The economy stays AC-only. Rewards climb from early hundreds to late thousands so upgrades feel meaningful without replacing scavenging, crafting, or BCP production.

Tier	Coverage	Mainline AC	Side AC	Puzzle AC	Notes
T1	Act I, B6-B1	200-450	250-500	150-450	Establish shelter, BCP basics, starter tool upgrades.
T2	Act II, B2-B3	450-900	500-950	350-900	Greenhouse, beacons, caravan logistics, first named weapons.
T3	Act III, B5-B4	900-1800	1000-2000	800-1800	Radiation/heat survival, advanced kits, faction conflicts.
T4	Act IV, B7/finales	1800-3500	2000-4500	1500-3000	Charter, endings, prestige gear, colony-scale choices.
Mastery	Optional puzzle mastery	N/A	N/A	600-3000	Prestige AC, cosmetics, weapon kits, and mastery tags.

4.2 Named Weapons And Upgrade Kits

Existing combat mapping: map named rewards to current combat/hotbar surfaces first: melee weapon stats, projectile/hotbar item data, stamina/tension costs, and inventory equip slots.

Future combat mapping: later systems can add unique animations, elemental behaviors, companion combo perks, and faction mod sockets without renaming the rewards.

Reward	Tier	Mapping	Source Examples
Horizon Shield	T1	Existing block/parry or defensive equip mapping; future Guard rally aura.	Horizon Guard SQs
Shelter Flare	T1	Hotbar consumable; future shelter path ping.	B1 storm quests
Lens Spade	T1-T2	Tool/harvest weapon; future agri node bonus.	Green Lens
Rivet Gun	T1-T2	Ranged/hotbar projectile mapping; future construction stagger.	Rebuild Guild
Beacon Rifle	T2	Existing ranged mapping; future waypoint tagging.	B3 Guard
Crane-Link	T2	Utility weapon/tool; future heavy object manipulation.	Rebuild Guild
Hearth Injector	T2	Hotbar support item; future companion triage perk.	Hearth Clinic
Mute Cart Rig	T2-T3	Stealth logistics tool; future convoy noise suppression.	Freeboot Relief
Radglass Carbine	T3	Ranged weapon; future radiation venting mod.	B5 puzzle/cache
Geothermal Maul	T3	Heavy melee mapping; future heat-charge swings.	B4 memorial forge
Vault-Cut Spear	T4	Melee reach weapon; future Precursor armor break.	B7 vault
Charter Sidearm	T4	Ranged prestige weapon; future ending-based mod.	C6/ending
Upgrade Kit: Field Mk I	T1	+baseline damage/durability.	Early sides/puzzles
Upgrade Kit: Shelter Mk II	T2	+survival utility, storm resistance, or reload safety.	B1-B3
Upgrade Kit: Resonance Mk III	T3	+core-linked effects, exposure tolerance.	B5-B4
Upgrade Kit: Horizon Mk IV	T4	+prestige stat package and ending visual.	B7/finales

5. Mainline Progress-Triggered Beat Sheet

Mainline beats advance through **progress triggers**, not quest givers. Triggers include locations reached, repairs completed, cores restored, biomes stabilized, buildings online, Echo milestones, and Resonance Events. NPCs can comment, but they do not hand out the spine.

Beat	Act	Trigger	Biome	Content	AC	Weapon / Kit	Core / Resonance	Trust Dialogue
ML-01 First Horizon	I	Player reaches B6 landing ridge and activates first BCP Overview.	B6	Establish hab shell, starter 5000 AC noted, assign first base worker.	200	Field Mk I kit	None	"One panel lit. That is how a horizon starts."
ML-02 Shelter Law	I	Command Center storm shelter comes online.	B6 -> B1	Storm shelter tutorial, base injury-not-death rule, B1 access.	250	Shelter Flare	None	"People bend in storms. Buildings must bend first."

Beat	Act	Trigger	Biome	Content	AC	Weapon / Kit	Core / Resonance	Trust Dialogue
ML-03 Coupler Repair	I	Habitat Power Coupler installed through BCP Changes.	B6	Repair object 1; Horizon Barrier pattern introduced.	300	Rivet Gun	Kairos stirs	"Do not build on their graves."
ML-04 Gyro Recovery	I	Storm Shelter Gyro recovered during B1 weather window.	B1	Repair object 2; Sulfur Hound pressure around shelter.	350	Horizon Shield	Kairos hostile night	"You came back before the peak. That matters."
ML-05 Warm-Boot Prism	I	Archive Warm-Boot Prism installed after B2 vent sync.	B2	Repair object 3; Kairos speaks clearly.	450	Lens Spade	Archive repair complete	"They were going to stay. Why did you come to finish dying?"
ML-06 Core 1: Founders' Roll	I	First Memory Core recovered from B1 roll vault.	B1	Save workers, read names, allied Echo phantoms aid harvest.	450	Shelter Mk II kit	C1 Resonance Beacon	"Read the roll aloud."
ML-07 Greenhouse Schema	II	Greenhouse module online in B2 Production tab.	B2	Fertility Mist event, crop trial, vent farming.	650	Nutrient Scanner	C2 Greenhouse Schema	"They mapped life onto poison."
ML-08 Beacon Homestead	II	Three B3 waystations powered.	B3	Scout Relay Mesh, caravan schedule, ash ridge map clarity.	800	Beacon Rifle	C3 Waystation Lights	"Homes need lights, not silence."
ML-09 Advisor Trust	II	Three cores/major Echo milestones restored.	B3	Kairos shifts from antagonist to advisor.	900	Field Mk II kit	Trust Stage: Advisor	"Ash gale building. Move harvest inside before it has a cost."
ML-10 Cold Promise	III	B5 habitat uplift plus rad suit installed.	B5	Polar camp recovery, Freeboot Relief arrival, Void Kelp survey.	1200	Radglass Carbine	C4 Warm Bubble	"Smugglers stole names. Steal them back."
ML-11 Hearth Of The Rim	III	B4 geothermal stabilizer attached to colony grid.	B4	Memorial forge, heat relay, grief-to-industry beat.	1600	Geothermal Maul	C5 Stability Bubble	"Mourn, then heat the forges."
ML-12 Charter Vault	IV	B7 Barrier Field opened by restored power grid.	B7	Three locks, precursor archive, C6 recovery.	2200	Vault-Cut Spear	C6 Charter Vault	"Write the next charter, or leave with the names."
ML-13 Charter Vote	IV	Six cores restored and BCP colony Overview reaches quorum.	B7 / Base	Charter UI opens: Build / Evacuate / Seed.	3000	Charter Sidearm	Final Resonance	"Hope is a system. Vote it into shape."

Beat	Act	Trigger	Biome	Content	AC	Weapon / Kit	Core / Resonance	Trust Dialogue
ML-14 Build Ending	IV	Player chooses Build, colony defenses and production online.	Base	Ratify permanent colony; Kairos becomes Colony Voice.	3500	Horizon Mk IV kit	Build resonance	"Then we stay, and we make staying worthy."
ML-15 Evacuate Ending	IV	Player chooses Evacuate, cores secured and launch route clear.	B7 -> B6	Leave with names, prevent erasure.	3500	Charter Mk IV kit	Evacuation archive	"Carry us where the sulfur cannot reach."
ML-16 Seed Ending	IV	Player chooses Seed, Purification Hub accepts Saturation cost.	UG5 / B7	Hybrid Echo-colony planted; ongoing legacy.	3500	Resonance Mk IV kit	Seed resonance	"Not ghosts. Not settlers. A third promise."

Mainline subtotal: 32h designed.

6. Side Quests - 40

Side and faction quests may use NPCs, faction boards, BCP work orders, companion prompts, and colony notices. AC follows the T1-T4 curve. Weapon rewards include named gear or upgrade kits where appropriate.

ID	Tier	Title	Biome	Giver / Board	Objectives	Ecology / Systems	Reward	Est.
SQ-V2-01	T1	Wall Raising	B6	Horizon Guard board	Place 3 cordon posts through BCP Changes and assign workers in Companions tab.	BCP shell [SHIPPED], modules [PLANNED]	300 AC, Horizon Shield	1.2h
SQ-V2-02	T1	Shelter Drill	B1	Horizon Guard	Move 4 workers to Command Center before storm Peak.	Weather stubs [PARTIAL], Sulfur Hound [SHIPPED]	350 AC, Shelter Flare x2	1.4h
SQ-V2-03	T1	Friendly Fire Check	B6	Horizon Guard / Hearth Clinic	Sync a frightened Echo while guards hold fire.	EchoGenerator [SHIPPED], Echo fail permanent	400 AC, Field Mk I kit	1.2h
SQ-V2-04	T2	Ridge Patrol	B3	Horizon Guard	Mark 5 waystations and repel Basalt Jackal ambushes.	Basalt Jackal [PLANNED], combat [SHIPPED]	700 AC, Beacon Rifle	1.3h
SQ-V2-05	T2	Barrier Power Share	B3	Horizon Guard	Route BCP Production power into a Horizon Barrier segment.	Barrier Field, BCP Craft/Changes [PARTIAL]	850 AC, Shelter Mk II kit	1.3h

ID	Tier	Title	Biome	Giver / Board	Objectives	Ecology / Systems	Reward	Est.
SQ-V2-06	T3	Guard Legend	B4	Horizon Guard	Escort memorial banners through Caldera Mantis territory.	Caldera Mantis [PLANNED], exposure [SHIPPED]	1600 AC, Guard Banner mod	1.8h
SQ-V2-07	T1	Seed Trial	B2	Green Lens Collective	Harvest during a Dissipating weather window and seed the first test bed.	Weather-as-schedule, Plume Moth [PLANNED]	450 AC, Lens Spade	1.2h
SQ-V2-08	T2	Nutrient Map	B2	Green Lens board	Sample 5 vent-fed nodes without overharvesting.	Vent Crabs, Dust Spout [PLANNED]	650 AC, Nutrient Scanner	1.1h
SQ-V2-09	T3	Polar Sprout	B5	Green Lens	Grow one cold crop under radiation pressure.	Void Kelp, exposure [SHIPPED]	1400 AC, Seedvault Case	1.5h
SQ-V2-10	T2	IP Or Open	B6	Green Lens / Helix	Choose whether greenhouse schema stays open-source or licensed.	Faction conflict, AC-only economy	750 AC, rep fork	1.0h
SQ-V2-11	T2	Bloom Night	B3	Green Lens	Harvest a Resonance bloom while Dust Spouts move across the ridge.	Resonance [PLANNED], Dust Spout	900 AC, Shelter Mk II kit	1.3h
SQ-V2-12	T4	Collective Finale	B7	Green Lens	Add a flora protection clause to the charter vote.	B7 [PLANNED], Charter UI [PLANNED]	2500 AC, Horizon Mk IV kit	1.5h
SQ-V2-13	T1	Hab From Wreck	B1	Rebuild Guild	Convert a wreck into a hab frame with Production and Changes tabs.	BCP [SHIPPED]/[PARTIAL]	450 AC, Rivet Gun	1.3h
SQ-V2-14	T2	Crane Rescue	B2	Rebuild Guild	Recover a Crane-Link from a geyser pit.	Vent timing, Heat Eel	700 AC, Crane-Link	1.2h
SQ-V2-15	T2	Scaffold War	B3	Rebuild Guild	Clear Tube Jackals from disputed scaffold piles without wrecking supplies.	Tube Jackal [PLANNED], combat [SHIPPED]	800 AC, Hab Patch kit	1.2h
SQ-V2-16	T3	Crust Foundation	UG3	Rebuild Guild	Mark safe brine crust for underground cellar farms.	Magnet Wyrms, UG [PLANNED]	1300 AC, Resonance Mk III kit	1.5h
SQ-V2-17	T2	Materials Court	B6	Rebuild / Caravan	Mediate access to salvage and common storage.	Quest scaffold [PARTIAL]	600 AC, rep fork	0.9h
SQ-V2-18	T3	Guild Legend	B4	Rebuild Guild	Build a geothermal memorial forge from a death-site.	Heat Eel, memorial pressure	1800 AC, Geothermal Maul	1.7h

ID	Tier	Title	Biome	Giver / Board	Objectives	Ecology / Systems	Reward	Est.
SQ-V2-19	T1	Print A Name	B6	Hearth Clinic	Reveal an Echo chronicle name and print a colony tag.	Echo chronicle [SHIPPED]	350 AC, Name Tag Printer	1.0h
SQ-V2-20	T1	Ryn's Hab	B6	Ryn / Hearth	Let Architect Ryn solve a habitability flaw in BCP Overview.	Companion trio, BCP shell	450 AC, Ryn bond, Field Mk I kit	1.3h
SQ-V2-21	T1	Nova's Storm Triage	B1	Nova / Hearth	Treat injured workers during storm Peak; no base deaths.	Exposure [SHIPPED], base injury-not-death	500 AC, Hearth Injector	1.4h
SQ-V2-22	T3	Calder's Care	B5	Calder / Hearth	Treat isotope exposure and choose public/private case notes.	Radiation, Void Kelp	1200 AC, Resonance Mk III kit	1.3h
SQ-V2-23	T2	Mira's Waylight	B3	Mira Storm / Hearth	Light a scout beacon chain without drawing Rift Stalkers.	Rift Stalker [PLANNED]	850 AC, Beacon charm	1.2h
SQ-V2-24	T2	Hub Debt	B6	Hearth Clinic	Cleanse 4 Saturation cases in Purification Hub prototype flow.	Purification Hub [PLANNED]	900 AC, Named Sync perk	1.3h
SQ-V2-25	T1	Commons Open	B6	Caravan Commons	Found the shared market and post first AC ledger.	AC [SHIPPED]	300 AC, Commons Pack	0.8h
SQ-V2-26	T1	Storm Schedule	B1	Caravan Commons	Deliver supplies before Peak or delay safely through shelter.	Weather stubs [PARTIAL]	450 AC, Schedule Slate	1.2h
SQ-V2-27	T2	Vent Caravan	B2	Caravan Commons	Escort freight across Dust Spout alley.	Dust Spout, Vent Crabs	650 AC, Shelter Mk II kit	1.2h
SQ-V2-28	T2	Ridge Freight	B3	Caravan Commons	Move freight past humanoid H2 raiders without losing crates.	H1-H3 humanoids [PLANNED]	850 AC, Shared Manifest perk	1.3h
SQ-V2-29	T3	Polar Chain	B5	Caravan Commons	Maintain a cold chain shipment to Green Lens crops.	Exposure, Void Kelp	1500 AC, Storm Ledger	1.2h
SQ-V2-30	T3	Commons Legend	B4	Caravan Commons	Escort heat-proof convoy to the memorial forge.	Caldera Mantis, Heat Eel	1900 AC, vendor stock unlock	1.6h
SQ-V2-31	T2	Continuity Offer	B2	Helix Continuity Mission	Accept, refuse, or restrict Helix funding for greenhouse IP.	External faction conflict	700 AC, Lease Stylus	0.8h
SQ-V2-32	T2	Audit Greenhouse	B2	Helix	Complete an audit without damaging live crops.	Greenhouse Schema [PLANNED]	900 AC, Audit Drone	1.2h
SQ-V2-33	T2	Echo As Asset	B6	Helix / Hearth	Decide whether Echo imprints can be classified as property.	Cores-are-people	950 AC, ending lean	1.1h

ID	Tier	Title	Biome	Giver / Board	Objectives	Ecology / Systems	Reward	Est.
SQ-V2-34	T2	Grant Gate Trial	B3	Helix	Fund one upgrade, then handle dependency strings.	BCP Changes [PARTIAL]	950 AC, Grant Gate perk	1.3h
SQ-V2-35	T4	Continuity Ultimatum	B7	Helix	Add, reject, or rewrite Helix charter terms.	Charter Vote [PLANNED]	3000 AC, Charter Sidearm mod	1.0h
SQ-V2-36	T3	Grey Seeds	B5	Freeboot Relief	Accept contraband seeds for sick workers.	Void Kelp, Freeboot conflict	1200 AC, Grey Seed Pouch	0.8h
SQ-V2-37	T1	Relief Run	B1	Freeboot Relief	Smuggle meds through a sulfur storm without Guard panic.	Sulfur Hound [SHIPPED], weather [PARTIAL]	500 AC, Relief Cloak	1.3h
SQ-V2-38	T2	Mute Cart	B3	Freeboot Relief	Deliver supplies past Guard sensors.	Barrier Field, H2 patrols	900 AC, Mute Cart Rig	1.3h
SQ-V2-39	T3	Barrier Mercy	B3	Freeboot / Guard	Open Horizon Field briefly for refugees.	Colony Barrier, faction conflict	1800 AC, Grey Bloom perk	1.4h
SQ-V2-40	T4	Freeboot Legend	B7	Freeboot Relief	Escort a charter witness past Vault Stalkers.	Vault Stalker, B7	4500 AC, Vault-Cut Spear	1.8h

Side quest subtotal: 52h designed.

7. Puzzle Quests - 40 Mixed Purpose

Required purpose distribution:

- **Access Gate:** 8, a minority.
- **Loot/Cache/Weapon:** 8.
- **Environmental Survival no gate:** 7.
- **Combat/Encounter:** 6.
- **Story/Reveal:** 6.
- **Optional Mastery:** 5.

ID	Tier	Purpose Type	Title	Biome	Solve Pattern	Ecology / Setpiece	Reward	Est.
PQ-V2-01	T1	Access Gate	Pad Field	B6	Route BCP power to a Horizon Barrier field.	Android A1 maintenance units	250 AC, Field Mk I kit	0.9h
PQ-V2-02	T1	Story/Reveal	Coupler Lattice	B6	Align habitat coupler symbols to expedition initials.	Kairos first grief fragments	300 AC, Ryn trust	1.1h

ID	Tier	Purpose Type	Title	Biome	Solve Pattern	Ecology / Setpiece	Reward	Est.
PQ-V2-03	T1	Environmental Survival no gate	Gyro Storm Lock	B1	Open wreck only between Peak and Dissipating storm phases.	Sulfur Hound patrol [SHIPPED]	350 AC, Shelter Flare	1.1h
PQ-V2-04	T2	Story/Reveal	Prism Vent Choir	B2	Sync three geyser tones to wake archive audio.	Plume Moth swarm, Vent Crabs	650 AC, Lens Spade mod	1.2h
PQ-V2-05	T1	Access Gate	Roll Vault	B1	Hold worker pressure plates while shelter siren cycles.	C1 Founders' Roll	450 AC, Shelter Mk II kit	1.2h
PQ-V2-06	T1	Loot/Cache/Weapon	Phantom Ally Cache	B1	Guide allied Echo phantoms to buried harvest crates.	EchoGenerator [SHIPPED]	400 AC, Horizon Shield mod	1.0h
PQ-V2-07	T2	Access Gate	Greenhouse Seals	B2	Balance contamination seals before agri module opens.	Heat Eel vent channels	750 AC, Nutrient Scanner	1.1h
PQ-V2-08	T2	Environmental Survival no gate	Fertility Valves	B2	Tune valves to avoid crop burn during Fertility Mist.	Plume Moth pollination hazard	700 AC, seed cache	1.0h
PQ-V2-09	T2	Access Gate	Waystation Triad	B3	Power three beacons in ridge order.	Basalt Jackal pressure	850 AC, Beacon Rifle	1.2h
PQ-V2-10	T3	Environmental Survival no gate	Warm Bubble Calibrator	B5	Balance cold/heat poles around a temporary habitat bubble.	Void Kelp radiation pockets	1100 AC, Radglass mod	1.1h
PQ-V2-11	T3	Access Gate	Lens Habitat Gate	B5	Pair rad pulse timing with habitat uplift.	Magnet Wyrms tremor intervals	1400 AC, Resonance Mk III kit	1.2h
PQ-V2-12	T3	Story/Reveal	Memorial Forge Lock	B4	Heat-lock a forge using recovered death-site names.	Caldera Mantis pressure	1700 AC, Geothermal Maul	1.3h
PQ-V2-13	T4	Access Gate	Charter Locks A	B7	Solve three surface locks tied to C1-C3 testimony.	Vault Stalker patrols	2200 AC, Vault-Cut Spear	1.5h
PQ-V2-14	T4	Access Gate	Charter Locks B	B7	Solve C4-C6 locks and prepare charter vote conditions.	Glass Hive resonance	2600 AC, Horizon Mk IV kit	1.5h
PQ-V2-15	T4	Story/Reveal	Charter Seal	B7	Align Build/Evacuate/Seed clauses with restored people.	Kairos final truth	3000 AC, Charter Sidearm	1.6h
PQ-V2-16	T3	Environmental Survival no gate	Gas Dome Farm	UG2	Vent gases into a safe agri mix.	Heat Eel, Tube Jackal	1200 AC, agri module kit	1.2h

ID	Tier	Purpose Type	Title	Biome	Solve Pattern	Ecology / Setpiece	Reward	Est.
PQ-V2-17	T3	Loot/Cache/Weapon	Dual Dome Cache	UG2	Sync two gas domes to uncover rare seed crates.	Plume Moth larvae	1300 AC, Seedvault upgrade	1.1h
PQ-V2-18	T3	Environmental Survival no gate	Crust Terrace	UG3	Mark safe brine crust without triggering collapse.	Magnet Wyrms tremors	1200 AC, cellar plan	1.2h
PQ-V2-19	T2	Loot/Cache/Weapon	Flooded Hydro Cache	UG3	Wade hydroponics route by pressure rhythm.	Void Kelp roots	900 AC, Hearth Injector cache	1.0h
PQ-V2-20	T2	Environmental Survival no gate	Rockfall Terrace	UG1	Drop controlled rockfall into future farm terraces.	Tube Jackal routes	800 AC, building materials	1.1h
PQ-V2-21	T2	Loot/Cache/Weapon	Guard Armory Code	B6	Decode Guard trust keypad after shelter drills.	Horizon Guard HQ	900 AC, Beacon Rifle mod	0.8h
PQ-V2-22	T2	Loot/Cache/Weapon	Cleanroom Bloom	B2	Sterilize grow beds in correct order.	Vent Crabs in ducts	850 AC, Lens Spade upgrade	1.0h
PQ-V2-23	T2	Loot/Cache/Weapon	Guild Crane Puzzle	B1	Reverse Crane-Link polarity to open a sealed crate.	Rebuild yard	800 AC, Rivet Gun upgrade	0.9h
PQ-V2-24	T2	Story/Reveal	Name Door	B6	Use printed Echo name tags as keys.	Hearth Clinic, Echo chronicle	750 AC, Named Sync record	0.9h
PQ-V2-25	T2	Loot/Cache/Weapon	Commons Freezer	B6	Re-power freezer through storm-safe grid.	Caravan Commons	900 AC, cold-chain kit	1.0h
PQ-V2-26	T2	Story/Reveal	Helix Lease Seal	B2	Break or validate lease clauses in a stylus cipher.	Helix Mission	950 AC, Continuity documents	1.1h
PQ-V2-27	T2	Loot/Cache/Weapon	Freeboot False Wall	B3	Move Mute Cart through an ash-cut wall.	Freeboot Relief	900 AC, Mute Cart Rig mod	1.0h
PQ-V2-28	T3	Access Gate	Barrier Grid	B3	Disable three linked Horizon Barrier fields.	Rift Stalker in field gaps	1600 AC, route unlock	1.4h
PQ-V2-29	T3	Combat/Encounter	Power Ethics	B3	Choose power siphon route while H3 raiders attack.	Humanoids H1-H3	1500 AC, Resonance Mk III kit	1.3h
PQ-V2-30	T2	Optional Mastery	Beacon Harmony	B3	Tune friendly frequencies without overloading beacons.	Silence Moth teaser	900 AC prestige, beacon cosmetic	1.1h
PQ-V2-31	T3	Environmental Survival no gate	Void Kelp Survey	B5	Scan live Void Kelp without exceeding exposure thresholds.	Void Kelp, Still Hunter shadow	1300 AC, rad filter	1.1h

ID	Tier	Purpose Type	Title	Biome	Solve Pattern	Ecology / Setpiece	Reward	Est.
PQ-V2-32	T2	Optional Mastery	Pulse Ride	B2	Ride geyser pulse cycles for a shortcut medal.	Dust Spout, flyers F1-F3	600 AC prestige, movement badge	1.0h
PQ-V2-33	T2	Optional Mastery	Spout Alley	B2	Cross a Dust Spout lane without shelter use.	Dust Spout	700 AC prestige, plume decal	1.0h
PQ-V2-34	T2	Combat/Encounter	Multi-Breach Hab	B3	Repair three breaches while Basalt Jackals attack.	Basalt Jackal	950 AC, Shelter Mk II kit	1.2h
PQ-V2-35	T3	Combat/Encounter	Cover Relay	B4	Rebuild cover relays during Caldera Mantis waves.	Caldera Mantis, Heat Eel	1600 AC, Geothermal mod	1.2h
PQ-V2-36	T4	Combat/Encounter	Android Rehab Cipher	B7	Reprogram or scrap android A8-A10 under fire.	Androids A8-A10, machines M6-M7	2200 AC, android frame	1.3h
PQ-V2-37	T4	Optional Mastery	Silent Refugee Path	B7	Mark a quiet route past Vault Stalkers with no alarm.	Vault Stalker, Silence Moth	2500 AC prestige, stealth charm	1.1h
PQ-V2-38	T3	Optional Mastery	Crust Lottery Farm	UG3	Solve a perfect safe lattice for bonus farm yield.	Magnet Wyrms, Rust Garden roots	3000 AC prestige, farm cosmetic	1.2h
PQ-V2-39	T3	Combat/Encounter	Dual Faction Barrier	B3	Use Guard and Freeboot tools while Rift Stalkers breach.	Rift Stalker	1800 AC, Grey Bloom perk	1.4h
PQ-V2-40	T4	Combat/Encounter	Legacy Cage	B7	Hold off Glass Hive and Void Stitcher while Evacuate seal opens.	Glass Hive, Void Stitcher	3000 AC, Charter Mk IV kit	1.4h

Puzzle subtotal: 45h designed.

8. Factions V2

Horizon Guard

- **Role:** Colony security, storm shelter discipline, wall rings, Barrier Field ethics.
- **Gear:** Horizon Shield, Beacon Rifle, Shelter Flare, Guard Banner mod.
- **Skills:** Shelter Rally, defensive regroup, escort discipline.
- **Quest identity:** Protect workers without turning the colony into a bunker.
- **Conflicts:** Freeboot Relief over smuggling and refugee access; Rebuild Guild over unsafe scaffolds; Helix over privatized security.

Green Lens Collective

- **Role:** Science, agriculture, vent greenhouse work, ecology observation.
- **Gear:** Lens Spade, Nutrient Scanner, Seedvault Case.
- **Skills:** Bloom Read, weather-window crop prediction, contamination reads.
- **Quest identity:** Turn poisonous landscapes into food systems without pretending lo is harmless.
- **Conflicts:** Helix wants schema ownership; Caravan Commons wants wider food access.

Rebuild Guild

- **Role:** Salvage-to-construction, hab conversion, BCP Changes tab work, module install.
- **Gear:** Rivet Gun, Crane-Link, Hab Patch kit.
- **Skills:** Fast Frame, salvage yield, structural repair.
- **Quest identity:** Builders learning to stop thinking like scrappers.
- **Conflicts:** Horizon Guard over safety; Caravan Commons over stockpiles.

Hearth Clinic

- **Role:** Medical care, Echo ethics, Purification Hub, no-base-death fiction.
- **Gear:** Hearth Injector, Name Tag Printer, Triage Cart.
- **Skills:** Named Sync, faster Echo stabilization when chronicle names are restored.
- **Quest identity:** People are not resources, including Memory Cores and Echoes.
- **Conflicts:** Helix classifies imprints as IP; Guard sometimes sees Echoes as threats.

Caravan Commons

- **Role:** Logistics, AC market ledgers, weather schedules, convoy work.
- **Gear:** Commons Pack, Schedule Slate, Storm Ledger.
- **Skills:** Shared Manifest, carry capacity, convoy timing.
- **Quest identity:** A colony becomes real when people trust the same ledger.
- **Conflicts:** Rebuild over materials priority; Guard over risky schedule windows.

Helix Continuity Mission

- **Role:** External corporate visitors, funding, contracts, continuity claims.
- **Gear:** Continuity Suit, Audit Drone, Lease Stylus.
- **Skills:** Grant Gate, funded upgrades with strings attached.
- **Quest identity:** Offers stability at the cost of ownership.
- **Conflicts:** Green Lens over greenhouse IP; Hearth Clinic over Echo personhood; final charter clauses.

Freeboot Relief

- **Role:** External smugglers turned aid runners, contraband medicine, refugee mercy.
- **Gear:** Relief Cloak, Mute Cart Rig, Grey Seed Pouch.
- **Skills:** Grey Bloom, quiet routes, nonstandard crop nodes.
- **Quest identity:** Law bends when people are outside the Barrier Field.

- **Conflicts:** Horizon Guard over access; Helix over unlicensed supplies; ally to Hearth Clinic.

9. Friction And Gate Matrix

Colony Horizon uses friction to schedule hope, not stall it. The best gates ask the player to build, shelter, route, repair, vote, or take responsibility.

ID	Friction / Gate	Primary Pattern	Emphasis	Content Using It
F1	Colony Barrier Field	Power route / BCP repair / faction access choice	Expansion corridors open when colony infrastructure earns them.	ML-03, ML-12, SQ-05, SQ-39, PQ-01, PQ-28
F2	Weather Window Cadence	Building -> Peak -> Dissipating -> Clear	Weather is a schedule; quests can be delayed, sheltered, or timed.	SQ-02, SQ-07, SQ-26, PQ-03, PQ-08
F3	Storm Shelter Mandate	Workers must reach Command Center during Peak.	Base injury-not-death; poor planning injures, does not kill base-22.	ML-02, SQ-02, SQ-21
F4	Building Control Panel Tabs	Overview, Companions, Production, Craft, Changes	Colony construction fantasy lives in in-world terminals.	ML-01, ML-07, SQ-01, SQ-13
F5	Echo Sync Stakes	Timed sync / name restore / permanent fail	Echo fail is permanent; retry means future content changes.	SQ-03, SQ-19, PQ-24
F6	Exposure Thresholds	Cold/heat/radiation load	B5/B4 do not only gate by item; they demand habitat planning.	ML-10, ML-11, SQ-09, PQ-10
F7	Gas Dome / Vent Farm	Valve timing and O2/agri mix	Survival puzzle with production payoff.	ML-07, PQ-16, PQ-17
F8	Brine Crust Risk	Safe path mapping / tremor timing	Optional survival and farm expansion, usually no hard gate.	SQ-16, PQ-18, PQ-38
F9	Strain / Saturation	Workload, Purification Hub, Echo cost	Hope has a maintenance cost, especially in Seed ending.	SQ-24, ML-16
F10	Memorial Pressure	Death-site respect / morale events	Old expedition deaths become colony practices, not loot flavor.	ML-11, SQ-06, SQ-18, PQ-12

10. Hour Budget

Designed hours exclude exploration, optional harvesting drift, base decoration, free roam, and post-ending sandbox.

Bucket	Hours
Mainline progress-triggered story	32
Side quests (40)	52
Puzzle quests (40)	45
Faction reputation overhead	12

Bucket	Hours
Systemic repeatables: shelter drills, weather windows, BCP production, Echo sync, Barrier reroutes, memorial events	29
Designed total excluding exploration	170
Exploration / free roam / building decoration bonus	25-45 bonus

Verdict: V2 Colony Horizon lands in the requested **165-175 designed hour** range and clears the 150h minimum without counting exploration.

11. Sample Dialogue

Kairos - Waking Hostile

"Do not route power through that wall. They sealed it while singing because the storm had already taken the roof. You do not get to call their graves infrastructure."

Kairos - First Trust Shift

"One name restored. One less person buried under a file hash. Keep going, builder."

Kairos - Advisor State

"Ash gale building west of Ridge 7. Move the harvest inside. Tell the Caravan Commons they have forty minutes if they want pride, twenty if they want people safe."

Kairos - Core 6

"The charter was never finished. They argued until the last shelter door closed. That is not failure. That is a colony trying to be honest before it was ready."

Horizon Guard

"A wall is not a threat. It is a promise that the worker behind it gets to sleep."

Green Lens Collective

"Nothing here is barren. It is just speaking in chemistry we have not earned yet."

Rebuild Guild

"Scrap is what you call a thing before somebody needs it. Give me a panel and a crew."

Hearth Clinic

"No more unnamed patients. Not Echoes, not cores, not the ones who wake up scared and useful."

Caravan Commons

"The ledger is not greed. It is how we prove the last crate went to a person, not a rumor."

Helix Continuity Mission

"We can protect your colony from collapse. All we ask is continuity of ownership, terminology, and future yield."

Freeboot Relief

"Call it smuggling when everyone is inside the field. Call it medicine when someone is still outside."

Build Ending

Kairos: "Then we stay. Not because I softened. Because you wrote rules strong enough to hold people through the next storm."

Evacuate Ending

Kairos: "Take the names. Leave the buildings if you must. A colony can fail and still save its people from erasure."

Seed Ending

Kairos: "Not ghosts. Not settlers. A third promise. Feed it carefully."

12. Phase 5 Legacy And Implementation Notes

- V2 should inherit Phase 5 legacy content as emotional archaeology, not as a competing canon layer.
- All economy text says **Aether Credits / AC** only.
- Journal Craft remains recipe library / scroll learning; production lives in Building Control Panels.
- Building Control Panels must remain central: Overview for status, Companions for assignments, Production for queues, Craft for recipes, Changes for modules and upgrades.
- Weather Director [PLANNED] is the major missing scheduler for storm shelter mandates and weather-window quest design.
- Kairos, six cores, Resonance Events, Communications, Purification Hub live behavior, biomes, underground, modules, and pet migration are [PLANNED].
- AC, EchoGenerator/chronicle, exposure, Sulfur Hound+B1 table, World Engine spine, BCP shell, combat, and hotbar are [SHIPPED] surfaces to build around.

- Weather stubs, BCP Craft/Changes, hovercraft, quest scaffold, and pets legacy are [PARTIAL] and should be connected carefully rather than treated as finished.